

ARPITHA V KHANAPUR

Robotics & Artificial Intelligence Student

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PROFESSIONAL SUMMARY

Final-year B.E. student in Robotics and Artificial Intelligence with hands-on experience in ROS, robot simulation, computer vision, machine learning, and embedded systems. Experienced in developing robotic simulations, sensorbased perception, and automation-oriented projects using ROS, Gazebo, RViz, Python, and C++.

TECHNICAL SKILLS

Programming: Python, C++, C, Java

Robotics: ROS1, ROS2, Gazebo, RViz, LiDAR, Robot Teleoperation

AI/ML: Machine Learning, CNN, Computer Vision, Digital Image Processing

Tools: Linux/Ubuntu, MATLAB, ANSYS, SolidWorks Autodesk/CAED, Arduino IDE

Core: Data Structures, Embedded Systems, Sensor Integration , PLC

EDUCATION

B.E. – Robotics and Artificial Intelligence

Expected 2027

Jawaharlal Nehru New College of Engineering, Shivamogga

Pre-University Education

2022

Deeksha Malnad PU College, Shivamogga

82%

High School

2020 Adhichunchangiri Shikshana Trust High School, Shivamogga 87.52%

EXPERIENCE

Robotics Simulation and Perception Engineer – Spaceborn

4 Months — Remote

WORKSHOPS & HANDS-ON EXPERIENCE

ARJUNA Robotics Workshop – Newrro Tech LLP, Bangalore

- Worked with the ARJUNA educational robotics platform using **ROS and Ubuntu**.
- Implemented `/cmd vel` teleoperation and visualized **LaserScan** data in RViz.

PROJECTS

Cleaning Robot Using ROS

- CurrentDeveloping an autonomous cleaning robot integrating **ROS, computer vision, and CNN-based waste classification**.
- Creating a custom dataset for vision-based object detection and classification. **Smart Retrofit – Conversion of Traditional Cars into Hybrids**
- Proposed an AI/ML-based system to analyze real-time engine signals and predict engine load and driving modes.
- Designed the concept to improve fuel efficiency without major hardware modifications. **Liquid Jaggery Automation**
- Designed an automation concept for liquid jaggery production using temperature monitoring and process control.
- Focused on improving process consistency and reducing manual intervention. **Machine Learning Projects**
- Developed projects including **Water Encroachment Detection, Customer Churn Prediction, Healthcare Chatbot, Handwritten Digit Recognition, Weather Prediction, and Movie Recommendation**.

ACHIEVEMENTS

- Won a quiz in **SUPER COMPUTERS** conducted by IISc Bangalore.

- Secured **3rd Prize** in the ML Gauntlet at TECHZONE college fest.
- Reached the finale of the **National Level Physical AI Robothon**.
- Secured **1st Prize** in a department-level poster presentation.
- Completed the **L&T Electric Vehicle course** with O Grade.

CERTIFICATIONS

- NEWRRO TECH LLP – ROS Certification
- L&T Edutech – Electric Vehicle Dynamics
- L&T Edutech – Fundamentals of Electric and Hybrid Electric Vehicle Technology
- My Equation – ROBO AI
- C, C++, Core Java & Python Programming Certification

SOFT SKILLS

Teamwork — Leadership — Communication — Problem Solving — Critical Thinking

LANGUAGES

English — Kannada — Hindi